

Assignment: Initramfs-Based Root Filesystem Selection

Objective

Design a system where the initramfs acts as the main application to select between two root filesystem partitions on an SD card during boot.

Description

The initramfs is used as the first user-space environment after the kernel boots. Instead of directly booting into a fixed root filesystem, the initramfs is responsible for controlling the boot process.

The initramfs initializes the system by mounting required virtual filesystems such as /proc, /sys, and /dev, and preparing device nodes.

After initialization, the initramfs prompts the user to choose between two available root filesystem partitions located on the SD card.

System Flow

1. Kernel boots and loads initramfs.
 2. Initramfs runs the initialization process.
 3. System mounts required virtual filesystems.
 4. Initramfs waits until SD card partitions are detected.
 5. A prompt is shown to the user to choose a partition.
 6. Based on user input, the selected partition is mounted.
 7. The system switches root from initramfs to the selected partition.
 8. Execution continues from the chosen root filesystem.
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System Structure

- Initramfs: contains minimal binaries and init script.
- Partition 1: first root filesystem.

- Partition 2: second root filesystem.